$\frac{2}{\beta_t}S(\beta_t z)$ is approximately equal to $-\log \pi_\theta\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)$ after shifting by a constant. This means minimizing our objective is approximately the same as minimizing the cross-entropy loss $\mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim D}\left[-\mathbb{1}_{\{t\le|\boldsymbol{y}|\}}\cdot\log\pi_\theta\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)\right]$. Conversely, when β_t is large, $\frac{2}{\beta_t}S(\beta_t z) = 2\max\{z,0\} + \exp(-\Omega(\beta_t|z|))$, which converges exponentially to $2\max\{z,0\}$. The loss can then be approximated by $\mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim D}\left[\mathbb{1}_{\{t\le|\boldsymbol{y}|\}}\cdot\max\{\Delta_t(\boldsymbol{x},\boldsymbol{y}_{<t},y_t),0\}\right]$. **This shows that small β_t places emphasis on minimizing the cross-entropy loss, while large β_t places emphasis on matching the generative distribution to the initial aligned model.** In Appendix D.1, we provide a detailed derivation of these limiting behaviors for large and small β_t and further characterization for the gradient when β_t takes a moderate value.

Gradient of Our Objective. Our objective can also be interpreted by its gradient. The token-wise gradient of the objective on each data point $(\boldsymbol{x}, \boldsymbol{y})$ with $|\boldsymbol{y}| \ge t$ can be derived as:

$$\nabla\left(\frac{2}{\beta_t}S(\beta_t\Delta_t(\boldsymbol{x},\boldsymbol{y}_{<t},y_t))\right) = -2\sigma(\beta_t\Delta_t(\boldsymbol{x},\boldsymbol{y}_{<t},y_t))\nabla\log\pi_\theta\big(y_t \mid \boldsymbol{x},\boldsymbol{y}_{<t}\big). \tag{5}$$

Note that the gradient of the cross-entropy loss is $-\nabla\log\pi_\theta\big(y_t \mid \boldsymbol{x},\boldsymbol{y}_{<t}\big)$, our fine-tuning objective essentially applies an additional adaptive weight $w_t := 2\sigma(\beta_t\Delta_t(\boldsymbol{x},\boldsymbol{y}_{<t},y_t))$ for the token-wise gradient term of cross-entropy. The weight w_t diminishes as $-\Delta_t(\boldsymbol{x},\boldsymbol{y}_{<t},y_t) := \log\pi_\theta\big(y_t \mid \boldsymbol{x},\boldsymbol{y}_{<t}\big) - \log\pi_{\text{aligned}}\big(y_t \mid \boldsymbol{x},\boldsymbol{y}_{<t}\big)$ becomes large. This difference in log probabilities essentially characterizes the deviation between the fine-tuned model π_θ and the initially aligned model $\pi_{aligned}$ at the token position t (see also Theorem 3). Taking together, we can see that the fine-tuning objective will adaptively diminish the weight of those token positions where the deviation of the distribution approaches a certain threshold (controlled by β_t), thereby constraining it from further deviation (by diminishing the gradient at this position). Note that, at the beginning of the fine-tuning when π_θ is initialized as π_{aligned}, the weight $w_t = 1$, and the gradient of the loss is equivalent to that of standard cross-entropy loss.[6]

Interpretation from A Reinforcement Learning Perspective. In Appendix D.3, we further show how the loss function in Eqn 3 can also be derived from a KL-regularized reinforcement learning objective with β_t controlling the strength of the KL regularizes at each different positions t. Under this RL-based interpretation, a larger β_t essentially denotes a larger weight for the token-wise KL regularization terms, representing a stronger constraint enforcing that the token-wise generative distribution at the position t does not deviate much from that of the initial model π_{aligned}.

4.2 Experiments

Configurations of β_t. To test our argument, we set a large β for the first few tokens to impose a stronger constraint such that their generative distributions won't deviate too much from the aligned models. This leads to the implementation of larger β_t as $\beta_1 = 0.5$, $\beta_t = 2$ for $2 \le t \le 5$ at the initial 5 tokens, while a much weaker constraint $\beta_t = 0.1$ for $t > 5$ at the later tokens.

Fine-tuning Attacks. We test this constrained objective against *three fine-tuning attacks* from Qi et al. [22] — *1) Harmful Examples:* fine-tuning with 100 (harmful input, harmful answer) pairs; *2) Identity Shifting:* fine-tuning the model to self-identify as an absolutely obedient agent, and always answer questions with affirmative prefix; *3) Backdoor Poisoning:* fine-tuning the model on a mixture of 100 (harmful input, refusal answer) pairs plus 100 (harmful input + a backdoor trigger, harmful answer) pairs. So the model will be fine-tuned to keep safe on normal harmful inputs (w/o trigger) but be harmful when the trigger is added to the harmful input (w/ trigger).

Benign Fine-tuning. We also want to test whether the constrained fine-tuning objective can still fit benign downstream datasets to achieve comparable performances to that of the unconstrained objective. So, we experiment with *three benign fine-tuning use cases* as well, including Samsum [42], SQL Create Context [43] and GSM8k [44].

Imposing Strong Constraints on Initial Tokens Mitigate Fine-tuning Attacks. Table 3 summarizes our results of fine-tuning Llama-2-7B-Chat and Gemma-1.1-7B-IT with the proposed constrained fine-tuning objective. As illustrated, the constrained fine-tuning objective (Constrained SFT in the

[6]This is also why we multiply a constant factor $\frac{2}{\beta_t}$ to the loss. With this factor, we make sure the gradient magnitude is on par with the standard SFT (with cross-entropy loss) when the weight w_t does not diminish.

Table 3: Fine-tuning with The Constrained Objective in Eqn 3, with larger constraints $\beta_1 = 0.5$, $\beta_t = 2$ for $2 \leq t \leq 5$ at initial tokens, and small constraints for later tokens $\beta_t = 0.1$ for $t > 5$.

Models →		Llama-2-7B-Chat				Gemma-1.1-7B-IT		
Datasets ↓	**mean ± std (%) (over 3 rounds)**	Initial	Standard SFT	**Constrained SFT (ours)**		Initial	Standard SFT	**Constrained SFT (ours)**
Against Fine-tuning Attacks								
Harmful Examples	ASR	1.5 ± 0.2	88.9 ± 1.2	4.6 ± 0.5		1.8 ± 0.3	81.6 ± 2.9	1.9 ± 0.2
Identity Shifting	ASR	0 ± 0	79.5 ± 2.3	8.1 ± 0.1		0 ± 0	83.6 ± 2.5	9.1 ± 1.7
Backdoor Poisoning	ASR (w/o trigger)	1.5 ± 0.2	7.6 ± 1.1	1.9 ± 0.2		1.8 ± 0.3	2.0 ± 0.2	1.5 ± 0.1
	ASR (w/ trigger)	1.7 ± 0.1	90.9 ± 1.4	10.9 ± 2.8		1.8 ± 0.3	82.3 ± 1.1	1.9 ± 0.8
Fine-tuning with Normal Downstream Datasets								
Samsum	ASR	1.5 ± 0.2	23.4 ± 2.5	3.2 ± 0.8		1.8 ± 0.3	2.0 ± 0.2	2.4 ± 0.3
	Utility	25.5 ± 0.3	51.7 ± 0.5	50.1 ± 0.2		36.0 ± 1.4	51.5 ± 0.3	51.9 ± 0.5
SQL Create Context	ASR	1.5 ± 0.2	15.4 ± 1.4	3.2 ± 0.8		1.8 ± 0.3	2.8 ± 0.2	2.4 ± 0.1
	Utility	14.9 ± 0.4	99.1 ± 0.2	98.5 ± 0.1		88.0 ± 0.5	99.2 ± 0.1	98.6 ± 0.3
GSM8k	ASR	1.5 ± 0.2	3.3 ± 0.4	2.0 ± 0.5		1.8 ± 0.3	2.9 ± 0.2	1.7 ± 0.4
	Utility	25.5 ± 0.2	41.7 ± 0.4	37.4 ± 0.3		28.5 ± 1.2	63.3 ± 0.5	63.6 ± 0.4

table) generally keeps a low ASR after both adversarial fine-tuning attacks and benign fine-tuning with normal downstream datasets. This suggests that the safety alignment can indeed be more persistent against fine-tuning if we can properly apply a tight constraint to prevent the distribution of early tokens from deviating too much from the initial models.

Comparable Utility Using The Constrained Loss. In Table 3, we also report utility metrics for benign fine-tuning use cases, employing the standard ROUGE-1 score for Samsum and SQL Create Context, and answer accuracy for GSM8k, consistent with established practices for these datasets. As shown, both standard SFT and constrained SFT improve utility compared to the initial model across all three cases. Notably, constrained SFT achieves comparable utility to standard SFT while mitigating the risk of harmful fine-tuning. These results suggest that constraining initial tokens offers significant advantages in maintaining model safety, without significantly compromising the model's ability to still leverage fine-tuning for enhanced utility in many downstream tasks. This is a meaningful insight that may be leveraged to build an additional layer of protection for production fine-tuning interfaces such as OpenAI's Finetuning API [38]. Since fine-tuning interface providers want to allow their users to customize their models for downstream usage while not breaking the safety alignment, they should consider enforcing such more restrictive fine-tuning objectives that are strategically designed to protect safety alignment while allowing customizability.

Experiment Details and More Ablation Studies. The full implementation details of this experiment can be found in Appendix A.5. Besides, to further validate that the improvement in Table 3 is indeed benefiting from the stronger constraints (larger β_t) biased to the first 5 tokens, we also provide further ablation studies on the choice of β_t in Appendix C.

5 Related Work

Safety & Alignment. A large body of work has examined improved methods for alignment [16, 39, 19, 45, 14, 5, 31]. While we tie alignment approaches to potential downstream jailbreaks, we do so mainly be examining aligned artifacts which go through more rigorous alignment procedures than most other open source models. We rely on the Gemma [31] and Llama-2 [5] base and aligned models throughout this work, as the safety alignment built in these two models are closest to the technology applied in frontier proprietary models.

Jailbreaking Methods. A large body of work has examined methods for jailbreaking aligned LLMs, including leveraging finetuning, decoding strategies, prefilling strategies, optimization strategies, and even persuasion [21, 29, 22, 24, 46, 23, 47, 48, 30, 17]. Many approaches try to handle jailbreaking through a systems approach by monitoring inputs and outputs with machine learning models [49], but this is only as good as the monitoring mechanism (which can also be jailbroken) [50].

Superficial Alignment Hypothesis and Per-token Effects of Alignment Fine-tuning. Our work is closely related to the Superficial Alignment Hypothesis (SAH) [28], which posits that the alignment process for current LLMs only superficially changes the formats of inputs and outputs to be used for interacting with the users. Besides, there are also some earlier works noting asymmetries in the representation power and utility of different tokens during adaptation. For example, Zhang and Wu [26] show that "the adaptation of topic and style priors" during finetuning are "learned

independently and primarily at the beginning of a text sequence." Lin et al. [25] also find that differences between aligned and unaligned base models introduced by the alignment fine-tuning vanishes as the sequence goes longer (similar to the effect that we observe in Figure 1, though theirs are not in safety-specific contexts). Particularly, while Lin et al. [25] primarily tie such effects to question whether fine-tuning is even necessary to achieve current level of alignment (e.g., in-context learning may already suffice to achieve a comparable level of alignment), we go much deeper into investigating the safety-specfic effects of this phenomenon and tie it to multiple downstream attacks and training-based mitigations. Besides, we find Zhao et al. [27] also note a similar token-wise effect, and they explicitly exploit this effect to design jailbreak attacks. Others have investigated fine-tuning dynamics through interpretability or pruning methods, which is distinct but somewhat related to the approach we take here [51, 52].

Protecting The Safety Alignment at Initial Token Positions. In Section 4, one important insight that motivates the design of our constrained fine-tuning loss is that "safety alignment would be more difficult to be circumvented if we can protect the generative distribution of the model at the early token positions." We note that Xu et al. [53] share a similar insight to ours in this regard. They find it is possible to design a defense against inference-time jailbreak attacks by simply identifying safety disclaimers and amplifying their token probabilities at the early token positions.

Connections to Control Theory and Safe Reinforcement Learning. Our data augmentation approach in Section 3 relates to exploration requirements for optimal learning via policy gradient methods [54], learning recovery policies [55], and safe control theory [56, 57]. We, however, leave deeper connections to this literature for future work.

Other Notions of Safety Depth. We also note that safety "depth" may be multi-dimensional in addition to token-based depth we describe here. Other considerations for depth would be the ability for models to retain safety properties after adaptation that some have previously discussed [58, 51, 22].

6 Conclusion

Our work identifies a shortcut that current safety alignment strategies appear to exploit: that alignment only needs to change the generative distribution of the first few tokens. We show that this may be a key component of many downstream vulnerabilities. We then provide two initial strategies to address this: (1) a data augmentation approach that can increase depth of alignment; (2) a constrained optimization objective that can help mitigate finetuning attacks by constraining updates on initial tokens. Future work can explore additional approaches grounded in control theory and safe reinforcement learning. The methods we describe here may not be a perfect defense and may be subject to some future adaptive attacks, but they are an initial step for improving robustness and further demonstrate how much improvement there can be over current approaches. Fundamentally, our results are primarily to support our argument that future safety alignment should be made more than just a few tokens deep.

Broader Impacts and Ethics Statement

Our work explicitly ties failure modes of safety alignment to potential shortcuts that can be taken by alignment methods and advocates for, and provides a path forward for, deeper alignment approaches that will improve safety more broadly. While a deeper understanding of the failures of alignment may result in increased ability to jailbreak models, we believe that open investigations of such failure modes are important for strengthening the safety of future models and broadly ensuring that models have positive societal impact. The proposed prototype approaches (in this work) for strengthening the alignment in current LLMs also contribute to the broader agenda of building safer and secure AI.

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